

Survival Analysis

Agenda



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B Kaplan-Meier Survival Curves and the Log-Rank Test

C The Cox Proportional Hazards Model and Its Characteristics

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Introduction to Survival Analysis

What is survival analysis?

Survival analysis is a collection of statistical procedures for data analysis for which the outcome **variable of interest is *time until an event occurs***.

Outcome variable: Time until an event occurs



Time-to-event data is common in many fields including, but not limited to:

- 1) Time from HIV infection to development of AIDS
- 2) Time to heart attack
- 3) Time to machine malfunction

The two most common questions related to survival analysis are:

- 1) What is the probability of survival to a certain point in time?
- 2) What is the average survival time?

Introduction to Survival Analysis

What is survival analysis?

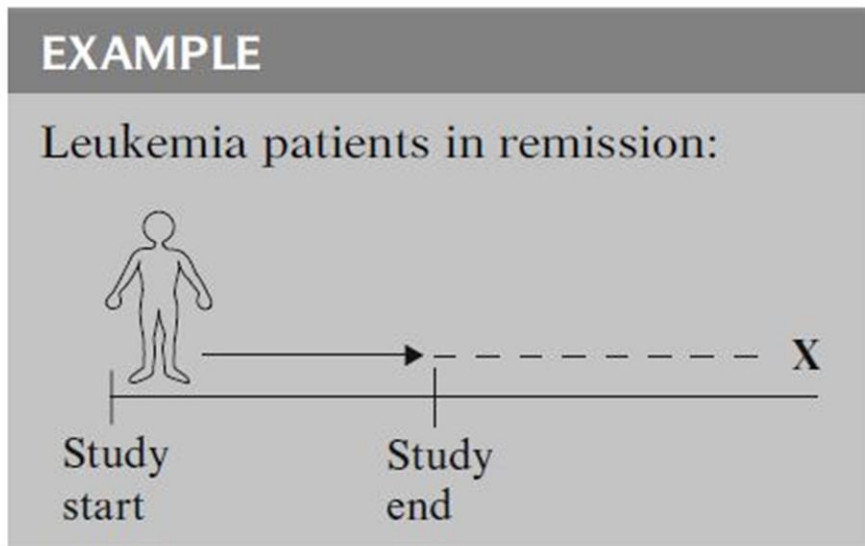
- Survival time can be measured in years, months, days, or even fractions of a second, alternatively, time can refer to the age of an individual ***when an event occurs***.
- By **event**, we mean death, disease incidence, relapse from remission, recovery (e.g., return to work) or any designated experience of interest that may happen to an individual.
- In a survival analysis, we usually refer to the **time variable** as survival time, because it gives the time that an individual has “survived” over some follow-up period.
- We also typically refer to the **event** as a failure, because the event of interest usually is death or some other negative individual experience

Introduction to Survival Analysis

Censored data

Key problem in survival analysis - **Censoring**

- It occurs when we have some information about individual survival time, but **we don't know the survival time exactly**.



$X \Rightarrow$ Event occurs

Introduction to Survival Analysis

Censored data

There are generally **three reasons** why censoring may occur:

(1) A person does not experience the event before the study ends;

(2) A person is lost to follow-up during the study period;

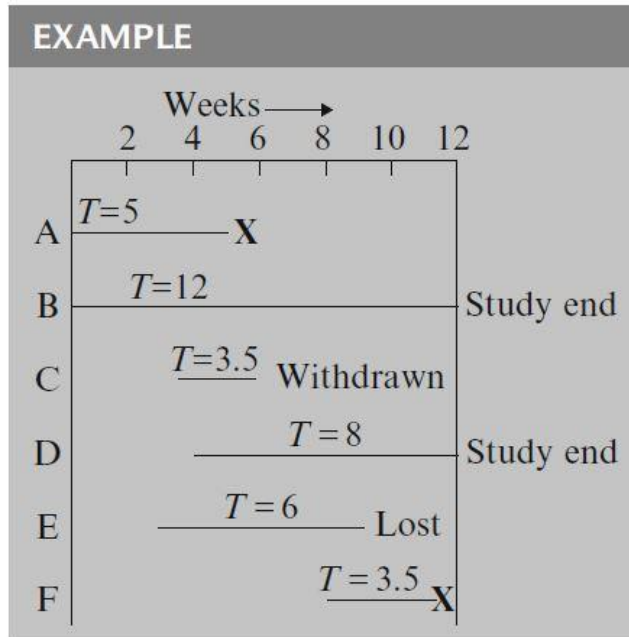
(3) A person withdraws from the study because of death (if death is not the event of interest) or some other reason (e.g., adverse drug reaction or other competing risk)



- **Censoring: don't know survival time exactly**

Introduction to Survival Analysis

Censored data



X $\Rightarrow$ Event occurs

Person **A** is followed from the start of the study until getting the event at week 5; **his survival time is 5 weeks** and is not censored.

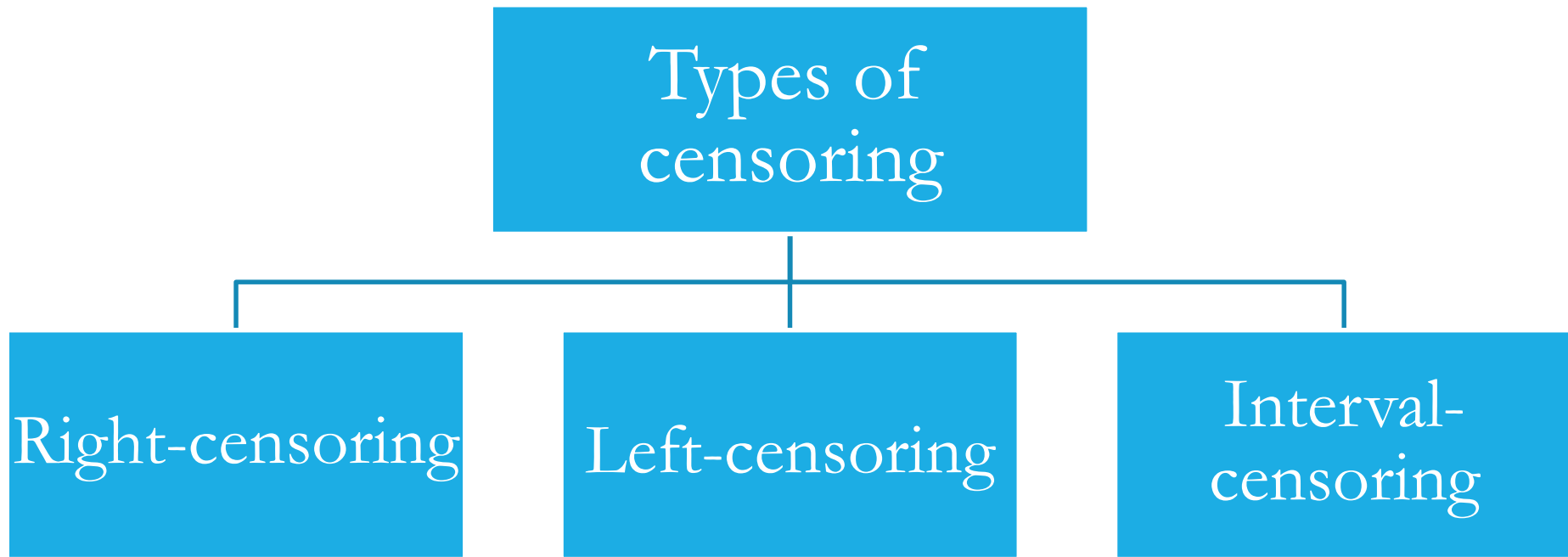
Person **B** also is observed from the start of the study but is followed to the end of the 12-week study period without getting the event; **the survival time here is censored** because we can say only that it is at least 12 weeks.

Person **C** enters the study between the second and 3rd week and is followed until he withdraws from the study at 6 weeks; **this person's survival time is censored** after 3.5 weeks.

In summary, of the six persons observed, two get the event (persons A and F) and four are censored (B, C, D, and E).

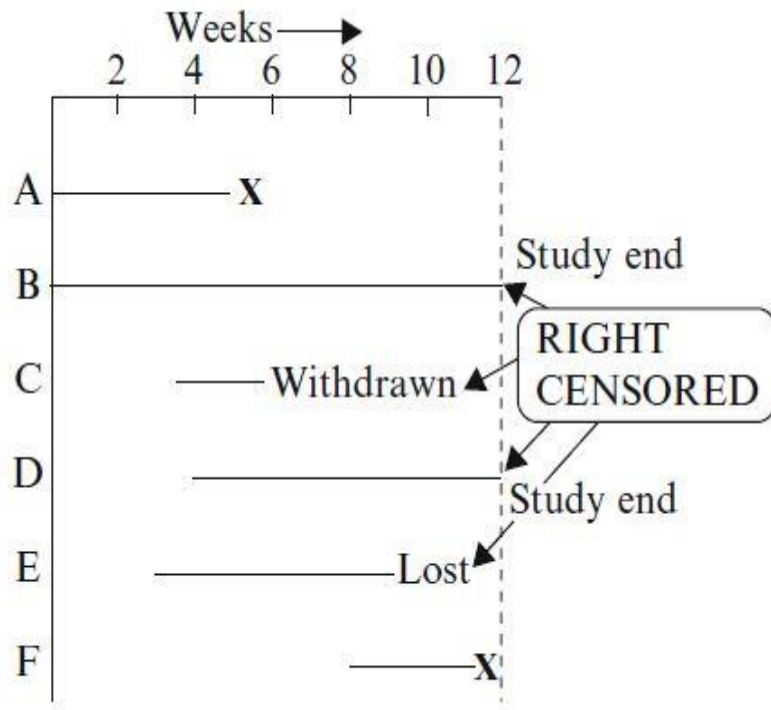
Introduction to Survival Analysis

Censored data



Introduction to Survival Analysis

Censored data - right



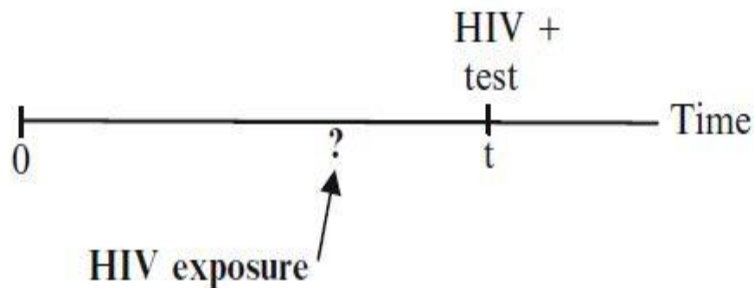
True survival time becomes incomplete at the right side of the follow-up period, occurring

1. when the study ends or
2. when the person is lost to follow-up or
3. is withdrawn.

Most survival data is right-censored.

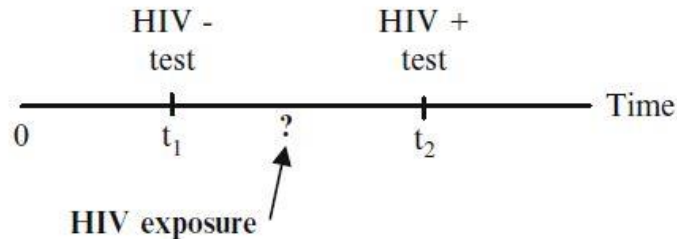
Introduction to Survival Analysis

Censored data – left; interval



Following persons until they become HIV positive, we may not know the exact time of first exposure to the virus, and therefore do not know exactly when the failure occurred.

The survival time is censored on the left side



Interval censored, can occur if a subject's true (but unobserved) survival time is within a certain known specified time interval.

Interval-censored: true survival time is within a known time interval

Kaplan-Meier Survival Curves and the Log-Rank Test

- The **Kaplan-Meier** curve illustrates the survival function. It's a step function illustrating the cumulative survival probability over time. The curve is horizontal over periods where no event occurs, then drops vertically corresponding to a change in the survival function at each time an event occurs.

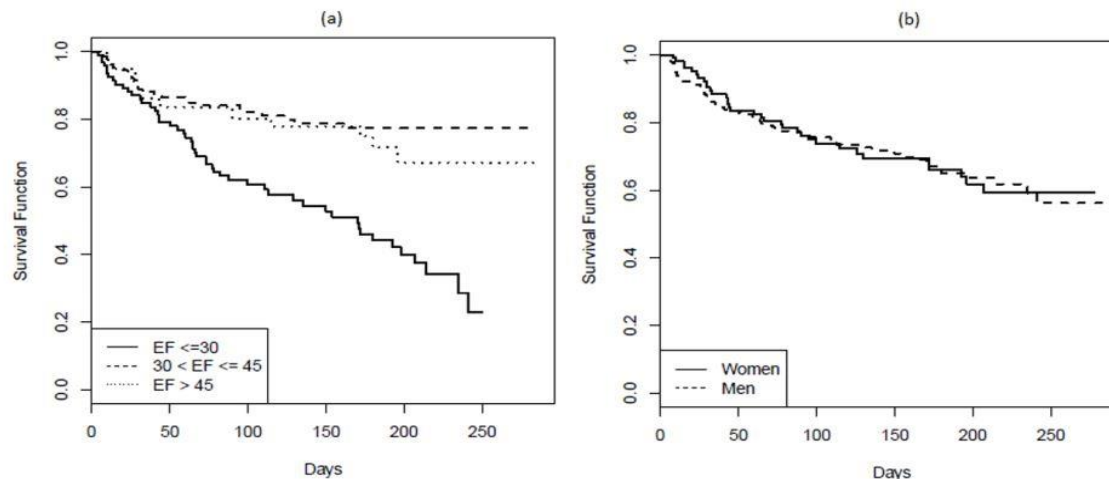
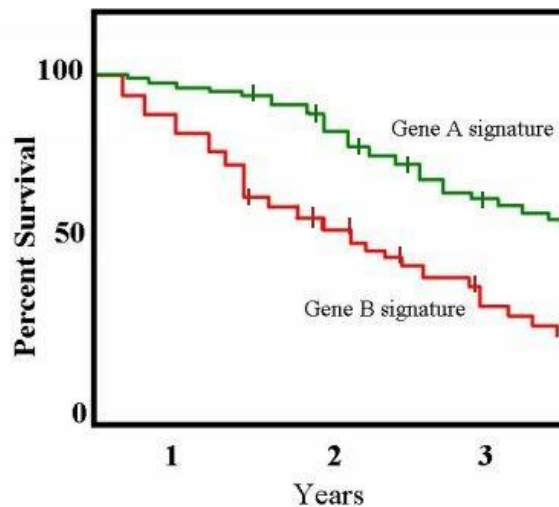


Fig 1. (a) Kaplan Meier curves were used to study the survival at different levels of EF and in part (b), survival curves of male and female.

Kaplan-Meier Survival Curves and the Log-Rank Test

It can be used to calculate:

- How long people remain unemployed after a job loss.
- How long it takes for couples undergoing fertility treatment to get pregnant.
- Time-to-failure of machine parts.
- In medicine, Kaplan Meier Analysis is the simplest way to calculate survival time after treatment.



This graph shows two groups of patients: one with gene profile A and one with gene profile B. People with gene B die at a faster rate than those with gene A.

Kaplan-Meier Survival Curves and the Log-Rank Test

For each time interval, survival probability is calculated as the number of subjects surviving divided by the number of patients at risk.

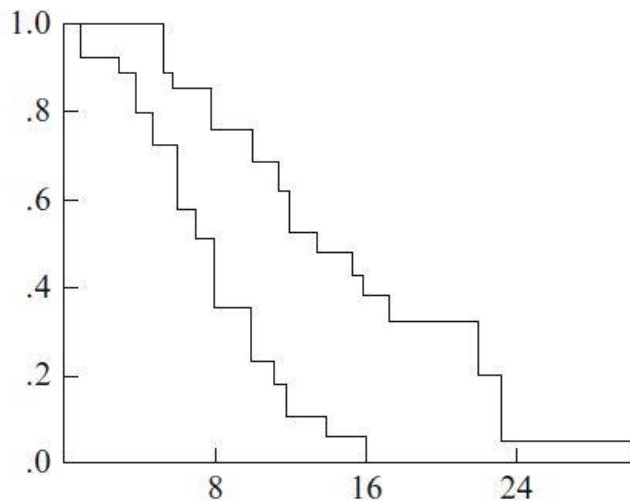
$$S_t = \frac{\text{Number of subjects living at the start} - \text{Number of subjects died}}{\text{Number of subjects living at the start}}$$

Total probability of survival till that time interval is calculated by multiplying all the probabilities of survival at all time intervals preceding that time.

E.g. **The probability of a patient surviving two days** after a kidney transplant can be considered to be probability of surviving the one day multiplied by the probability surviving the second day given that patient survived the first day.

Kaplan-Meier Survival Curves and the Log-Rank Test

Are KM curves statistically equivalent?



When we state that two KM curves are “statistically equivalent,” we mean that, based on a testing procedure that compares the two curves in some “overall sense,” we do not have evidence to indicate that the true (population) survival curves are different.

Log-rank test, tests the null hypothesis that the survival curves in the two groups are the same.

The log-rank test is the most widely used method of comparing two or more survival curves. The null hypothesis is that there is no difference in survival between the two groups. The log rank test is a non-parametric test, which makes no assumptions about the survival distributions. Essentially, the log rank test compares the observed number of events in each group to what would be expected if the null hypothesis were true (i.e., if the survival curves were identical). The log rank statistic is approximately distributed as a chi-square test statistic.

The Cox Proportional Hazards Model and Its Characteristics

The purpose of the model is to evaluate simultaneously the effect of several factors on survival. It allows us to examine how specified factors influence the rate of a particular event happening (e.g., infection, death) at a particular point in time. This rate is commonly referred as the **hazard rate**.

$$h(t) = h_0(t) \times \exp(b_1x_1 + b_2x_2 + \dots + b_px_p)$$

where,

- t represents the survival time
- $h(t)$ is the hazard function determined by a set of p covariates $(x_1, x_2, \dots, x_p)$
- the coefficients $(b_1, b_2, \dots, b_p)$ measure the impact (i.e., the effect size) of covariates.
- the term h_0 is called the baseline hazard. It corresponds to the value of the hazard if all the x_i are equal to zero (the quantity $\exp(0)$ equals 1). The 't' in $h(t)$ reminds us that the hazard may vary over time.

The Cox Proportional Hazards Model and Its Characteristics

Kaplan-Meier curves are good for visualizing differences in survival between two categorical groups, but they don't work well for assessing the effect of *quantitative* variables like age, leukocyte count, etc.

Cox PH regression can assess the effect of both categorical and continuous variables, and can model the effect of multiple variables at once.

Why Cox PH model is so popular?

It is a “robust” model, so that the results from using the Cox model will closely approximate the results for the correct parametric model.

It is preferred over the logistic model when survival time information is available and there is censoring. Because you can obtain more information!

Implementation in R

Install and load required R package

We'll use two R packages:

- *survival* for computing survival analyses
- *survminer* for summarizing and visualizing the results of survival analysis

Install the packages

```
install.packages(c("survival", "survminer"))
```

Load the packages

```
library("survival")  
library("survminer")
```

Implementation in R

Example data sets

We'll use the lung cancer data available in the survival package.

```
data("lung")  
head(lung)
```

	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
1	3	306	2	74	1	1	90	100	1175	NA
2	3	455	2	68	1	0	90	90	1225	15
3	3	1010	1	56	1	0	90	90	NA	15
4	5	210	2	57	1	1	90	60	1150	11
5	1	883	2	60	1	0	100	90	NA	0
6	12	1022	1	74	1	1	50	80	513	0

inst: Institution code

time: Survival time in days

status: censoring status 1=censored, 2=dead

age: Age in years

sex: Male=1 Female=2

ph.ecog: ECOG performance score
(0=good 5=dead)

ph.karno: Karnofsky performance score
(bad=0-good=100) rated by physician

pat.karno: Karnofsky performance score as
rated by patient

meal.cal: Calories consumed at meals

wt.loss: Weight loss in last six months

Implementation in R

Compute survival curves: `survfit()`

The function `survfit()` [in *survival* package] can be used to compute kaplan-Meier survival estimate. Its main arguments include:

- a survival object created using the function `Surv()`
- and the data set containing the variables.

We use lung cancer data; To compute survival curve, type:

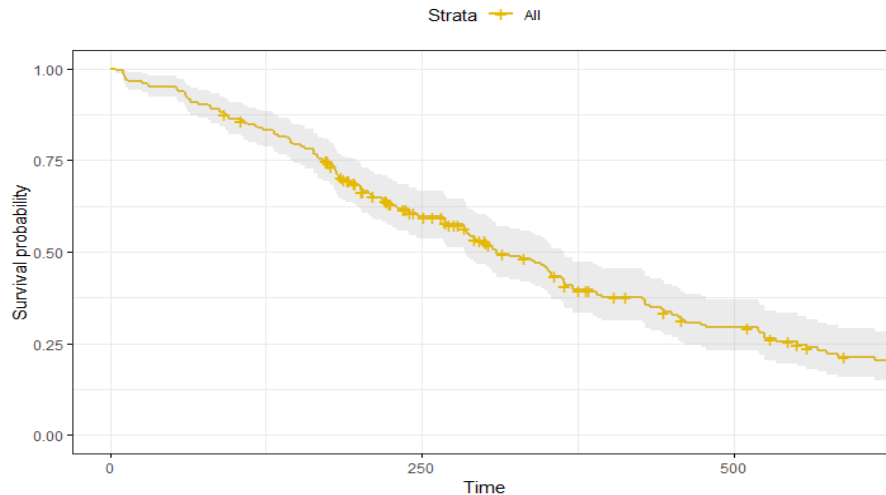
```
sfit <- survfit(Surv(time, status) ~ 1, data=lung)
```

To plot survival curve, type:

```
ggsurvplot(sfit,  
  conf.int = TRUE,  
  risk.table.col = "strata", # Change risk table color by groups  
  ggtheme = theme_bw(), # Change ggplot2 theme  
  palette = c("#E7B800", "#2E9FDF"),  
  xlim = c(0, 600))
```

Implementation in R

Compute survival curves: survfit()



The median survival time can be obtained using the code below:

```
summary(fit)$table
```

records	n.max	n.start	events	*rmean	*se(rmean)
228.00000	228.00000	228.00000	165.00000	376.27475	19.70779
median	0.95LCL	0.95UCL			
310.00000	285.00000	363.00000			

✓ The median survival time represent the time at which the survival probability, $S(t)$, is 0.5.

The median survival time for the people with lung cancer is 310days.

Implementation in R

Compute survival curves: `survfit()`



We want to compute the survival probability by sex.

The function `survfit()` [in *survival* package] can be used to compute kaplan-Meier survival estimate. Its main arguments include:

- a survival object created using the function `Surv()`
- and the data set containing the variables.

To compute survival curves, type:

```
fit <- survfit(Surv(time, status) ~ sex, data = lung)
print(fit)
```

```
Call: survfit(formula = Surv(time, status) ~ sex, data = lung)
      n events median 0.95LCL 0.95UCL
sex=1 138    112   270    212    310
sex=2  90     53   426    348    550
```



By default, the function `print()` shows a short summary of the survival curves. It prints the number of observations, number of events, the median survival and the confidence limits for the median.

Implementation in R

Access to the value returned by `survfit()`

The function *survfit()* returns a list of variables, including the following components:

- **n**: total number of subjects in each curve.
- **time**: the time points on the curve.
- **n.risk**: the number of subjects at risk at time t
- **n.event**: the number of events that occurred at time t.
- **n.censor**: the number of censored subjects, who exit the risk set, without an event, at time t.
- **lower,upper**: lower and upper confidence limits for the curve, respectively.
- **strata**: indicates stratification of curve estimation. If strata is not NULL, there are multiple curves in the result. The levels of strata (a factor) are the labels for the curves.

Implementation in R

Access to the value returned by survfit()

The components can be accessed as follow:

```
d <- data.frame(time = fit$time,  
                n.risk = fit$n.risk,  
                n.event = fit$n.event,  
                n.censor = fit$n.censor,  
                surv = fit$surv,  
                upper = fit$upper,  
                lower = fit$lower  
                )
```

```
head(d)
```

	time	n.risk	n.event	n.censor	surv	upper	lower
1	11	138	3	0	0.9782609	1.0000000	0.9542301
2	12	135	1	0	0.9710145	0.9994124	0.9434235
3	13	134	2	0	0.9565217	0.9911586	0.9230952
4	15	132	1	0	0.9492754	0.9866017	0.9133612
5	26	131	1	0	0.9420290	0.9818365	0.9038355
6	30	130	1	0	0.9347826	0.9768989	0.8944820

Implementation in R

Visualize survival curves

We'll use the function `ggsurvplot()` [in *Survminer* R package] to produce the survival curves for the two groups of subjects.

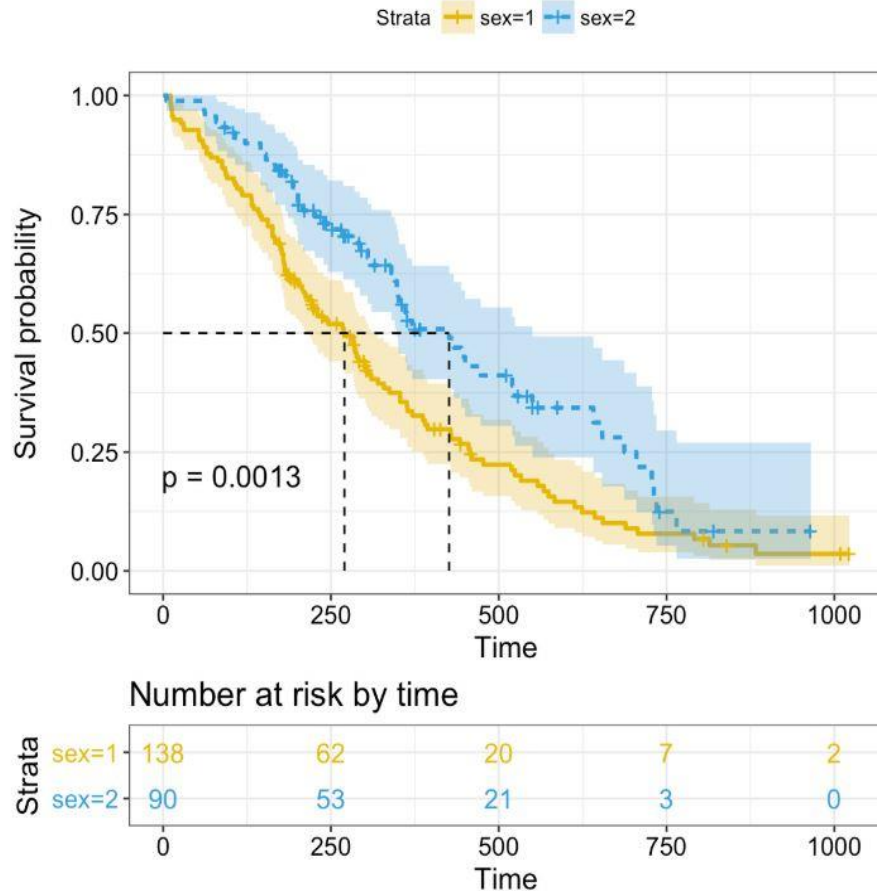
```
ggsurvplot(fit,
  pval = TRUE, conf.int = TRUE,
  risk.table = TRUE, # Add risk table
  risk.table.col = "strata", # Change risk table color by groups
  linetype = "strata", # Change line type by groups
  surv.median.line = "hv", # Specify median survival
  ggtheme = theme_bw(), # Change ggplot2 theme
  palette = c("#E7B800", "#2E9FDF"))
```

Implementation in R

Visualize survival curves

The Kaplan-Meier plot can be interpreted as follow:

The horizontal axis (x-axis) represents time in days, and the vertical axis (y-axis) shows the probability of surviving or the proportion of people surviving. The lines represent survival curves of the two groups. A vertical drop in the curves indicates an event. The vertical tick mark on the curves means that a patient was censored at this time.



- At time zero, the survival probability is 1.0 (or 100% of the participants are alive).
- At time 250, the probability of survival is approximately 0.55 (or 55%) for sex=1 and 0.75 (or 75%) for sex=2.
- The median survival is approximately 270 days for sex=1 and 426 days for sex=2, suggesting a good survival for sex=2 compared to sex=1

Implementation in R

Visualize survival curves

The median survival times for each group can be obtained using the code below:

```
summary(fit)$table
```

	records	n.max	n.start	events	*rmean	*se(rmean)	median	0.95LCL	0.95UCL
sex=1	138	138	138	112	325.0663	22.59845	270	212	310
sex=2	90	90	90	53	458.2757	33.78530	426	348	550

✓ The median survival times for each group represent the time at which the survival probability, $S(t)$, is 0.5.

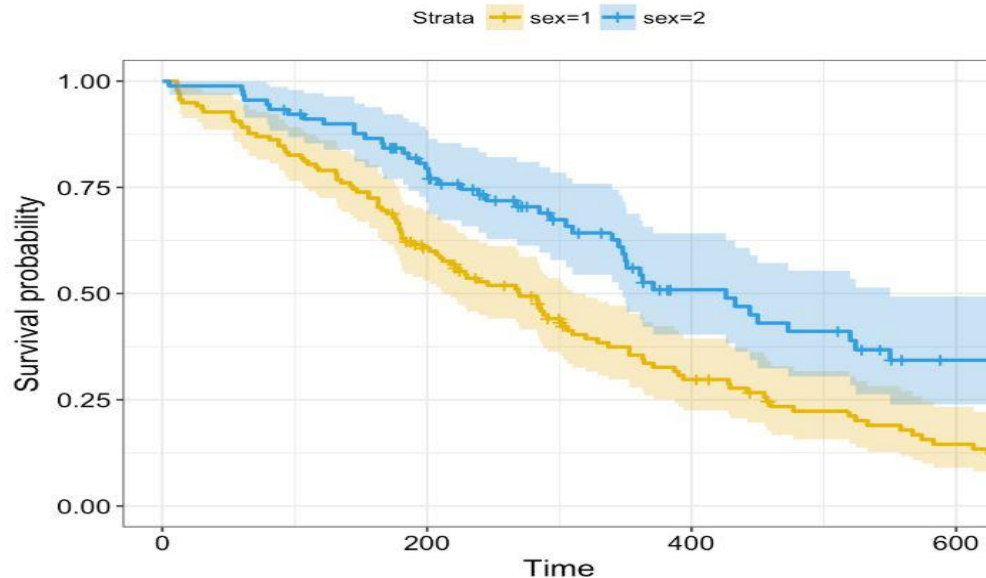
The median survival time for sex=1 (Male group) is 270 days, as opposed to 426 days for sex=2 (Female). There appears to be a survival advantage for female with lung cancer compare to male. However, to evaluate whether this difference is statistically significant requires a formal statistical test, a subject that is discussed in the next sections.

Implementation in R

Visualize survival curves

The survival curves can be shortened using the argument *xlim* as follow:

```
ggsurvplot(fit,  
  conf.int = TRUE,  
  risk.table.col = "strata", # Change risk table color by groups  
  ggtheme = theme_bw(), # Change ggplot2 theme  
  palette = c("#E7B800", "#2E9FDF"),  
  xlim = c(0, 600))
```

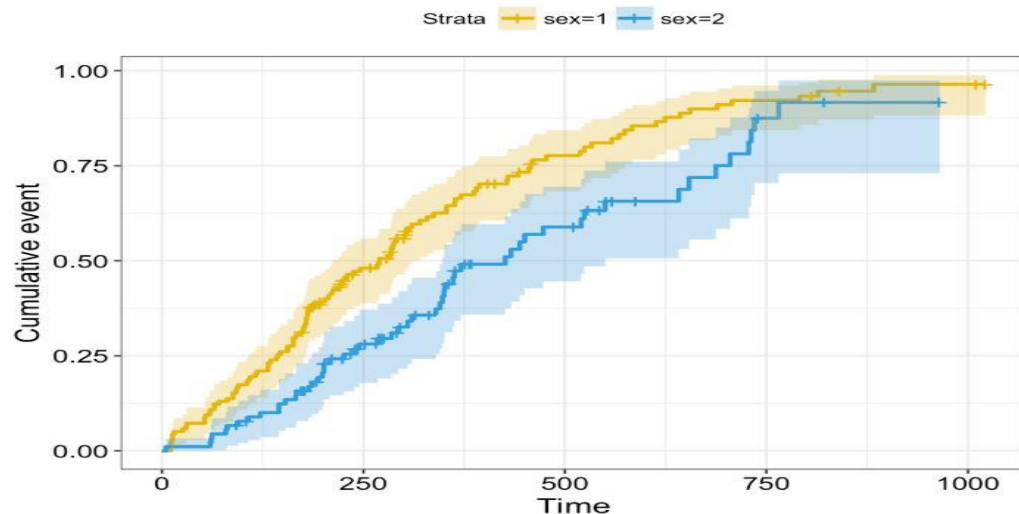


Implementation in R

Visualize survival curves

To plot cumulative events, type:

```
ggsurvplot(fit,
  conf.int = TRUE,
  risk.table.col = "strata", # Change risk table color by groups
  ggtheme = theme_bw(), # Change ggplot2 theme
  palette = c("#E7B800", "#2E9FDF"),
  fun = "event")
```



Note that, three often used transformations can be specified using the argument *fun*:

- "log": log transformation of the survivor function,
- "event": plots cumulative events ($f(y) = 1 - y$). It's also known as the cumulative incidence,
- "cumhaz" plots the cumulative hazard function ($f(y) = -\log(y)$)

Implementation in R

Kaplan-Meier life table: summary of survival curves

The function **survdif()** [in *survival* package] can be used to compute log-rank test comparing two or more survival curves.

Use the function **surv_summary()** [in *survminer* package] to get a summary of survival curves.

```
res.sum <- surv_summary(fit)
head(res.sum)
```

	time	n.risk	n.event	n.censor		surv	std.err	upper	lower	strata	sex
1	11	138	3	0	0.9782609	0.01268978	1.0000000	0.9542301	sex=1	1	
2	12	135	1	0	0.9710145	0.01470747	0.9994124	0.9434235	sex=1	1	
3	13	134	2	0	0.9565217	0.01814885	0.9911586	0.9230952	sex=1	1	
4	15	132	1	0	0.9492754	0.01967768	0.9866017	0.9133612	sex=1	1	
5	26	131	1	0	0.9420290	0.02111708	0.9818365	0.9038355	sex=1	1	
6	30	130	1	0	0.9347826	0.02248469	0.9768989	0.8944820	sex=1	1	

The function **surv_summary()** returns a data frame with the following columns:

- **time:** the time points at which the curve has a step.
- **n.risk:** the number of subjects at risk at t.
- **n.event:** the number of events that occur at time t.
- **n.censor:** number of censored events.
- **surv:** estimate of survival probability.
- **std.err:** standard error of survival.
- **upper:** upper end of confidence interval
- **lower:** lower end of confidence interval
- **strata:** indicates stratification of curve estimation. The levels of strata (a factor) are the labels for the curves.

Implementation in R

Log-Rank test comparing survival curves: `survdif()`

`survdif()` can be used as follow:

```
surv_diff <- survdiff(Surv(time, status) ~ sex, data = lung)
surv_diff
```

```
Call:
survdif(formula = Surv(time, status) ~ sex, data = lung)
      N Observed Expected (O-E)^2/E (O-E)^2/V
sex=1 138      112     91.6      4.55      10.3
sex=2  90       53     73.4      5.68      10.3
Chisq= 10.3 on 1 degrees of freedom, p= 0.00131
```

The function returns a list of components, including:

- **n**: the number of subjects in each group.
- **obs**: the weighted observed number of events in each group.
- **exp**: the weighted expected number of events in each group.
- **chisq**: the chisquare statistic for a test of equality.
- **strata**: optionally, the number of subjects contained in each stratum.



The log rank test for difference in survival gives a p-value of $p = 0.0013$, indicating that the sex groups differ significantly in survival.

Implementation in R

Cox Regression

Cox PH regression can assess the effect of both categorical and continuous variables, and can model the effect of multiple variables at once. The `coxph()` function uses the same syntax as `lm()`, `glm()`, etc. The response variable you create with `Surv()` goes on the left hand side of the formula, specified with a `~`. Explanatory variables go on the right side

Let's go back to the lung cancer data:

```
fit <- coxph(Surv(time, status)~sex, data=lung)
fit
```

Call:

```
coxph(formula = Surv(time, status) ~ sex, data = lung)
```

	coef	exp(coef)	se(coef)	z	p
sex	-0.531	0.588	0.167	-3.18	0.0015

```
Likelihood ratio test=10.6 on 1 df, p=0.00111
n= 228, number of events= 165
```

The `exp(coef)` column is the **hazard ratio** – the multiplicative effect of that variable on the hazard rate (for each unit increase in that variable).

So, for a categorical variable like sex, going from male (baseline) to female results in approximately ~40% reduction in hazard. You could also flip the sign on the coef column, and take `exp(0.531)`, which you can interpret as being male resulting in a 1.7-fold increase in hazard, or that males die at approximately 1.7x the rate per unit time as females (females die at 0.588x the rate per unit time as males).

Implementation in R

Cox Regression

Just remember:

HR=1: No effect

HR>1: Increase in hazard

HR<1: Reduction in hazard (protective)

You'll also notice there's a p-value on the sex term, and a p-value on the overall model. That 0.00111 p-value is really close to the $p=0.00131$ p-value we saw on the Kaplan-Meier plot.

That's because the KM plot is showing the log-rank test p-value.

Implementation in R

Cox Regression

Let's create another model where we analyze all the variables in the dataset!
This shows us how all the variables, when considered together, act to influence survival.

```
fit <- coxph(Surv(time, status)~sex+age+ph.ecog+  
ph.karno+pat.karno+meal.cal+wt.loss, data=lung)  
fit
```

Call:

```
coxph(formula = Surv(time, status) ~ sex + age + ph.ecog + ph.karno +  
      pat.karno + meal.cal + wt.loss, data = lung)
```

	coef	exp(coef)	se(coef)	z	p
sex	-5.51e-01	5.76e-01	2.01e-01	-2.74	0.0061
age	1.06e-02	1.01e+00	1.16e-02	0.92	0.3591
ph.ecog	7.34e-01	2.08e+00	2.23e-01	3.29	0.0010
ph.karno	2.25e-02	1.02e+00	1.12e-02	2.00	0.0457
pat.karno	-1.24e-02	9.88e-01	8.05e-03	-1.54	0.1232
meal.cal	3.33e-05	1.00e+00	2.60e-04	0.13	0.8979
wt.loss	-1.43e-02	9.86e-01	7.77e-03	-1.84	0.0652

```
Likelihood ratio test=28.3 on 7 df, p=0.000192  
n= 168, number of events= 121  
(60 observations deleted due to missingness)
```

Some are very strong predictors (sex, ECOG score). Interestingly, the Karnofsky performance score as rated by the physician was marginally significant, while the same score as rated by the patient was not.

Survival Analysis to Explore Customer Churn

Churn Prediction in Mobile Social Games: Towards a Complete Assessment Using Survival Ensembles

Churn Prediction in Mobile Social Games: Towards a Complete Assessment Using Survival Ensembles

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Abstract—Reducing user attrition, i.e. churn, is a broad challenge faced by several industries. In mobile social games, decreasing churn is decisive to increase player retention and rise revenues. Churn prediction models allow to understand player loyalty and to anticipate when they will stop playing a game. Thanks to these predictions, several initiatives can be taken to retain those players who are more likely to churn.

Survival analysis focuses on predicting the time of occurrence of a certain event, churn in our case. Classical methods, like regressions, could be applied only when all players have left the game. The challenge arises for datasets with incomplete churning information for all players, as most of them still connect to the game. This is called a censored data problem and is in the nature of churn. Censoring is commonly dealt with survival analysis techniques, but due to the inflexibility of the survival statistical algorithms, the accuracy achieved is often poor. In contrast, novel ensemble learning techniques, increasingly popular in a variety of scientific fields, provide high-class prediction results.



The work presented in this paper focuses on predicting churn for *high value players who are commonly called whales* in the video game industry.

Survival Analysis to Explore Customer Churn

A motivation for focus on predicting churn for *high value players* is that whales behave differently than average players.

Since *whales* are often the most active players, i.e. they play nearly every day, we can easily define their churn as a prolonged period of inactivity. Their high level of engagement also allows to collect more data about their activity and makes them more likely to answer positively to actions taken in order to prevent their churn.

Dataset contains mainly game-independent features, i.e. features that are not related to the game mechanics and can be measured in any game (e.g. *Time spent per day in the game*, *Number of days with at least one playing session*, *Days since last purchase*).

Conventional churn prediction is solved from a static point of view, a binary classification problem.

Our work focuses on *when churn will happen*.

Survival Analysis to Explore Customer Churn

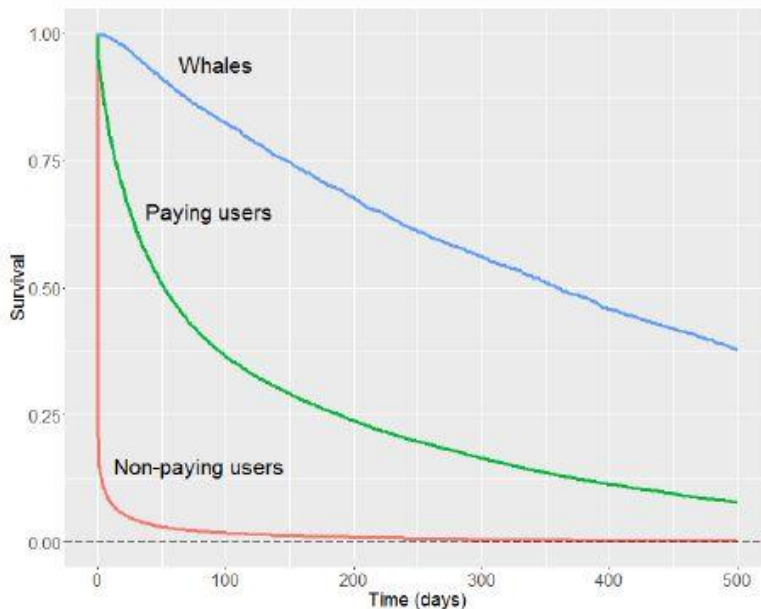


Fig. 2. Cumulative survival probability (Kaplan-Meier estimates) as a function of time, in days. The whole dataset is considered, and the results are stratified by whales (high-value players), paying users, and non-paying users.

Fig. 2 shows different survival patterns for each group. The estimated survival for non-paying users is much lower than the one of paying users (including both whales and non-whales paying users).

Approximately 80% of the non-playing players have churned the first day they connected to the game.

This contrasts with the 20% churn rate of the whales after 100 days.

Survival Analysis to Explore Customer Churn

Player Retention In League of Legends: A Study Using Survival Analysis

Player Retention In League of Legends: A Study Using Survival Analysis

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ABSTRACT

Multi-player online esports games are designed for extended durations of play, requiring substantial experience to master. Furthermore, esports game revenues are increasingly driven by in-game purchases. For esports companies, the trends in players leaving their games therefore not only provide information about potential problems in the user experience, but also impacts revenue. Being able to predict when players are about to leave the game - churn prediction - is therefore an important solution for companies in the

KEYWORDS

League of Legends, churn prediction, game analytics, esports, prediction, business intelligence, churn

ACM Reference format:

Simon Demediuk, Alexandra Murrin, David Bulger, Michael Hitchens, Anders Drachen, William L. Raffe, and Marco Tamassia. 2018. Player Retention In League of Legends: A Study Using Survival Analysis. In *Proceedings of Australasian Computer Science Week 2018, Brisbane, QLD, Australia, January 29-February 2, 2018 (ACSW 2018)*, 9 pages.

The objective of the work presented here is to understand the impact of specific behavioral characteristics on the likelihood of a player continuing to play the esports title League of Legends.

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Analysis is based on behavioral characteristics of players, namely Recent Average Match Duration (RAMD) and Recent Average Time Between Matches (RATBM), both being calculated over a 90 days period prior to the last event. Also Highest Achieved Season Tier (HAST) is used as a proxy for the skill level of the player. The response variable is Time Until Next Match (TUNM).

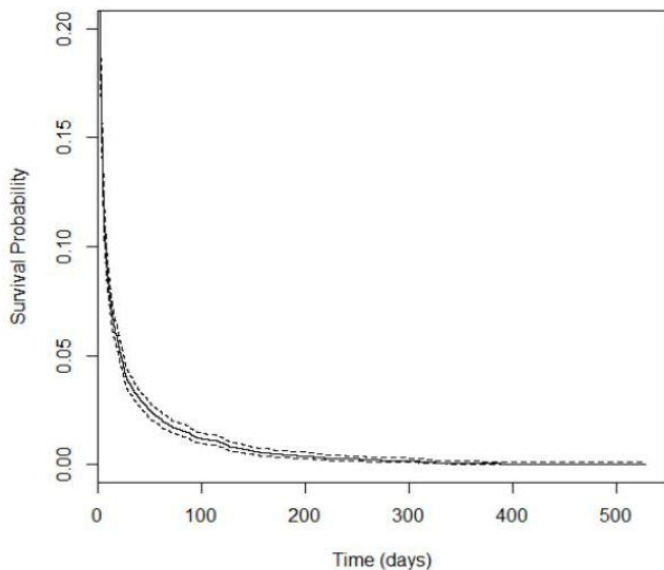


Figure 5: Kaplan-Meier Survival Function for Time Until Next Match with 95% confidence intervals.

Figure 5 depicts a decreasing rate of decline in survival probability over time after the initial severe drop.

This indicates that as the time between a match increases the probability of a another consecutive match being played decreases.

The curve begins to plateau around day 100 indicating that players with Time Until Next Match greater than 100 days have very low probability of returning to the game.

Survival Analysis to Explore Customer Churn

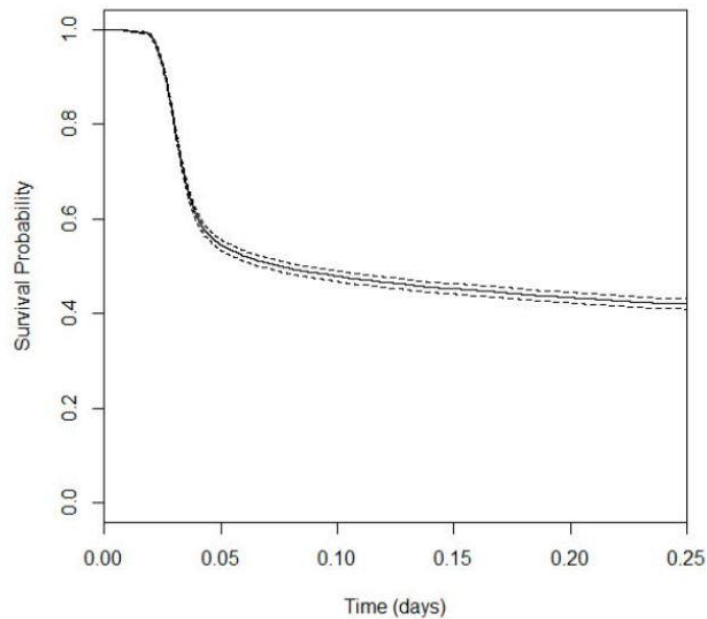


Figure 6: Kaplan-Meier Survival Function for Time Until Next Match with 95% confidence intervals over a quarter-day period.

Given the sharp initial decline, the survival function is again plotted in Figure 6 over a period of 6 hours (0.25 days) given 58% of observations are below this threshold.

It is easier to observe that the probability of the observations consisting of a match not yet having been played declines to around 50% after approximately 1 hours and 12 minutes (0.05 of a day).

Summary

Survival analysis is a set of statistical approaches for data analysis where the outcome variable of interest is time until an event occurs.

Survival data are generally described and modeled in terms of two related functions:

- The survivor function representing the probability that an individual survives from the time of origin to some time beyond time t . It's usually estimated by the Kaplan-Meier method. The logrank test may be used to test for differences between survival curves for groups, such as treatment arms.
- The hazard function gives the instantaneous potential of having an event at a time, given survival up to that time. It is used primarily as a diagnostic tool or for specifying a mathematical model for survival analysis.

Survival Analysis